

Find a Real Problem

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can describe a specific user difficulty using supplied evidence without inventing its cause.

I can define a useful outcome and a way to observe it before choosing a solution.

I can choose a small project with clear boundaries, permitted information and a simple alternative.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

A strong project begins with a specific user and observable need.



LEARNING TARGET Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

Learner name:

Readiness for the connected explanation

A teacher permits a paper prototype using fictional activity requests. Does this permit collecting real pupils' names and publishing their choices? Explain.

This week's end goal

By the end, I can turn a recorded problem into a bounded, measurable project goal.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY 'Students need AI' is not a problem statement; 'students spend five minutes finding misplaced supplies after art' is observable and testable.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

For the final project sequence, we will develop a small activity recommender on paper. Its users are fictional pupils choosing a classroom activity that fits their available time and room conditions. We will decide what the recommender must do, build a traceable process, test it and explain its limits. A recommendation is a suggestion; the pupil can decline it.

Start before the solution. Our supplied observation log describes ten fictional requests made using an activity catalogue. Six first choices fit the request's time and room conditions. Four did not and needed another choice. These invented classroom records give us a problem to investigate. They are not observations of your classmates and do not show how often this happens in other classes.

A useful problem statement is: "In the supplied ten-request exercise, four first choices did not fit the stated

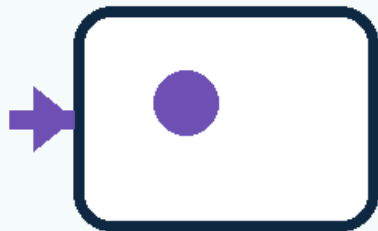
time or room conditions, so those users had to choose again.” “We need AI” names a possible technology but says neither who needs help nor what should improve. The log also does not establish whether unclear wording, missing information or another cause produced the mismatches.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



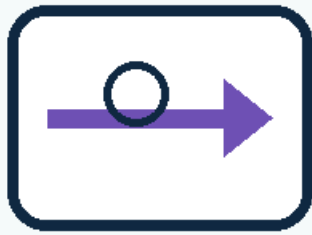
INPUT

observations and interview evidence



PROCESS

identify user, situation, need, and measurable effect



OUTPUT

testable problem statement without a
preselected solution

INPUT observations and interview evidence

PROCESS identify user, situation, need, and measurable effect

OUTPUT testable problem statement without a preselected solution

CORE IDEA A strong project begins with a specific user and observable need.

UNDER THE HOOD Problem framing separates evidence about a user's need from assumptions about a solution. A strong statement identifies who, when, what observable difficulty occurs, and how improvement could be measured.

Draw a fresh example with all three stages:

A closer explanation

From an observed difficulty to a testable goal

Evidence or decision · What it supports · What remains to check

Evidence or decision: 4 of 10 choices mismatch

What it supports: Some re-selection occurred

What remains to check: Cause is not established

Evidence or decision: Need a compatible choice

What it supports: Check time and conditions

What remains to check: User can still decline

Evidence or decision: Paper suggestion process

What it supports: Small permitted prototype

What remains to check: Benefit needs a comparison

Evidence or decision: Measure fit and coverage

What it supports: Observe useful outcomes

What remains to check: Do not hide no-result cases

The goal describes a useful result. Building a tool is an activity, not evidence that users benefited.

Our target outcome is a compatible suggestion when the catalogue contains one, and a clear no-match message when none fits. Compatibility means respecting the stated conditions; it does not mean that every pupil will enjoy the activity. We can check time and room conditions from a request and catalogue. Enjoyment would need different evidence.

Count compatible suggestions, incompatible suggestions and no-match messages separately. Otherwise a system could avoid every wrong suggestion by refusing to suggest anything. If a valid option existed, that refusal would still fail to provide the intended help. Record the total number of requests so a count has a clear denominator.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

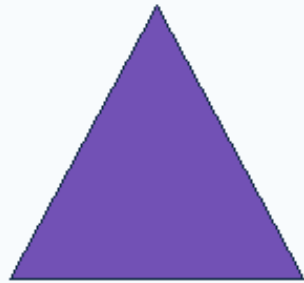
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



Students need AI



Books are messy



Librarian re-sorts 20 books daily



One student opinion

CASE 1 — Students need AI Clue: technology-first claim. Meaning: weak framing.

CASE 2 — Books are messy Clue: vague judgment. Meaning: needs observation.

CASE 3 — Librarian re-sorts 20 books daily Clue: specific burden. Meaning: testable need.

CASE 4 — One student opinion Clue: useful but limited. Meaning: seek more evidence.

BOUNDARY Do not collect personal stories unrelated to the design question.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: The approved list contains no tray with every requested tool.

Prediction: The finder should report no suitable tray rather than guess.

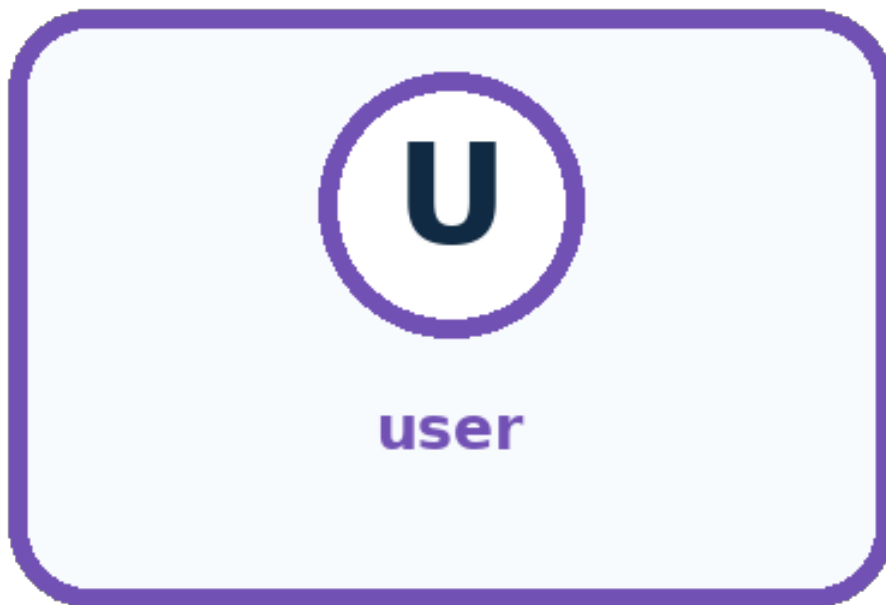
Reason: Success includes honestly identifying unsatisfied requirements.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





need



interview



Word · Meaning · My example

Word: user

Meaning: person affected by the problem or solution

My example: _____

Word: need

Meaning: observable difficulty or goal

My example: _____

Word: interview

Meaning: structured questions used to learn from a person

My example: _____

Word: success measure

Meaning: evidence showing whether the need improved

My example: _____

Misconception detective

CLAIM TO REPAIR A popular technology is proof of a real problem.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: User need

Meaning: Something a person needs to accomplish in a particular situation

Word: Problem statement

Meaning: A specific description of who faces a difficulty and what that difficulty is

Word: Outcome

Meaning: The result experienced by the user

Word: Constraint

Meaning: A condition that a proposed solution must respect

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE 'Students need AI' is not a problem statement; 'students spend five minutes finding misplaced supplies after art' is observable and testable.

STEP 1 — OBSERVE observations and interview evidence

STEP 2 — TRACE identify user, situation, need, and measurable effect

STEP 3 — CONCLUDE testable problem statement without a preselected solution

MODEL REASONING A strong statement names the user, context, observable difficulty, and desired improvement without requiring AI. Protect privacy, avoid leading questions, and distinguish what the user said from your inference.

DO NOT OVERCLAIM Do not collect personal stories unrelated to the design question.

What new evidence could change this answer?

A closer explanation

Project choice · Boundary for this sequence · Reason

Project choice: Users and setting

Boundary for this sequence: Fictional classroom activity requests

Reason: A clear situation to test

Project choice: Information

Boundary for this sequence: Available minutes and quiet-room requirement

Reason: Relevant to compatibility

Project choice: Action

Boundary for this sequence: Suggest an activity or report no match

Reason: The user retains the choice

Project choice: Outside the project

Boundary for this sequence: Names, profiles, purchases and public posting

Reason: Not needed for this paper task

The teacher permits a paper prototype using fictional requests and an approved catalogue. We do not need names, home details or a history of a pupil's choices. A request can say "10 minutes, quiet room required" without identifying anyone. Use only the request information needed to test the intended outcome.

A clear catalogue or a short checklist may solve this problem. We will compare a proposed recommender with a simple starting method instead of assuming a more complicated system is better. This project can use explicit rules; it does not have to fit a machine-learning model. The choice should follow the problem, available evidence and limits of the task.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Find a Real Problem

A fictional class loses time finding the correct craft tray.

The facts we will use

In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

Find a Real Problem

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Observe problem

Three mismatches are recorded out of eight requests.



2. Define need

Help users choose a tray containing every requested tool.



3. Set success test

Compare selected tray against the approved inventory for each new case.



4. Bound the project

Use fictional tool requests; do not collect identity or purchase equipment.

CHANGE ONE THING

The approved list contains no tray with every requested tool.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Observe problem

What happens and why: Three mismatches are recorded out of eight requests.

Step: 2. Define need

What happens and why: Help users choose a tray containing every requested tool.

Step: 3. Set success test

What happens and why: Compare selected tray against the approved inventory for each new case.

Step: 4. Bound the project

What happens and why: Use fictional tool requests; do not collect identity or purchase equipment.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The mismatch record describes a concrete need. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it. Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

What this example does not establish

Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

Read the labelled picture: Write a one-sentence project goal and say what evidence would count as a failure.

Change one thing: The approved list contains no tray with every requested tool. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: Students need AI

Evidence: technology-first claim

What it shows: weak framing

Case: Books are messy

Evidence: vague judgment

What it shows: needs observation

Case: Librarian re-sorts 20 books daily

Evidence: specific burden

What it shows: testable need

Case: One student opinion

Evidence: useful but limited

What it shows: seek more evidence

Repair two tempting claims

CLAIM A A popular technology is proof of a real problem.

Repair:

CLAIM B The first interview represents every user.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

Before building, say what would count against your idea. If a clear catalogue already produces compatible choices for all supplied cases, adding a recommender might offer little benefit on those cases. If the recommender hides valid options or produces unexplained no-match messages, revise it or keep the simpler method. A project can succeed as an investigation even when the evidence favours a simpler solution.

Keep the ten-request observation log separate from later evaluation records. It motivates the problem; it is not a fair test result for a tool that has not been built. Later weeks will introduce explicit requirements and test cases, including a reserved final set. Label plans as plans and measured results as results.

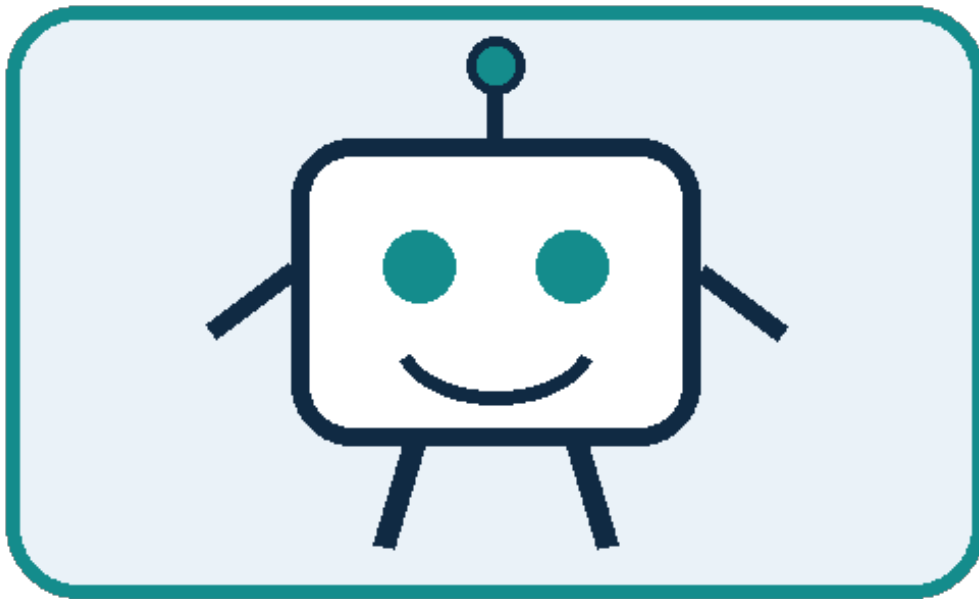
A proposed system always says “Ask the teacher”. It never suggests an incompatible activity. Has this alone demonstrated the intended benefit? Explain which additional result you would need to check

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A strong project begins with a specific user and observable need.

Use a fresh example to explain user.

Trace a new input → process → output example.

Find and repair the error in: The first interview represents every user.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Suppose a team writes “All pupils hate the catalogue because it is too long.” The supplied log says nothing about every pupil’s feelings or the cause of the mismatches. A more careful note is “Four of ten first

choices were incompatible; the cause is not yet known.” A hypothesis such as “the time information may be hard to find” is something to investigate, not an established finding.

Write a short project brief for the activity problem. Include the users and setting, the supplied evidence, the desired outcome, information needed, one boundary and a simple alternative. Name one claim the ten-request log does not establish

If we later investigate the cause, we could inspect where compatibility information appears in the catalogue or have a teacher organise a permitted comparison of clearer formats. Do not quietly collect real pupils’ data just because it might help. For this sequence, the supplied fictional records are enough to build and test a paper process.

Practise with less help: Identify claims beyond the evidence

Use workbook W2 for the connected example “A problem worth solving”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for A problem worth solving. Proposed explanation: “Using AI is itself the outcome”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Write a one-sentence project goal and say what evidence would count as a failure.

My explanation using a pictured step and the given facts

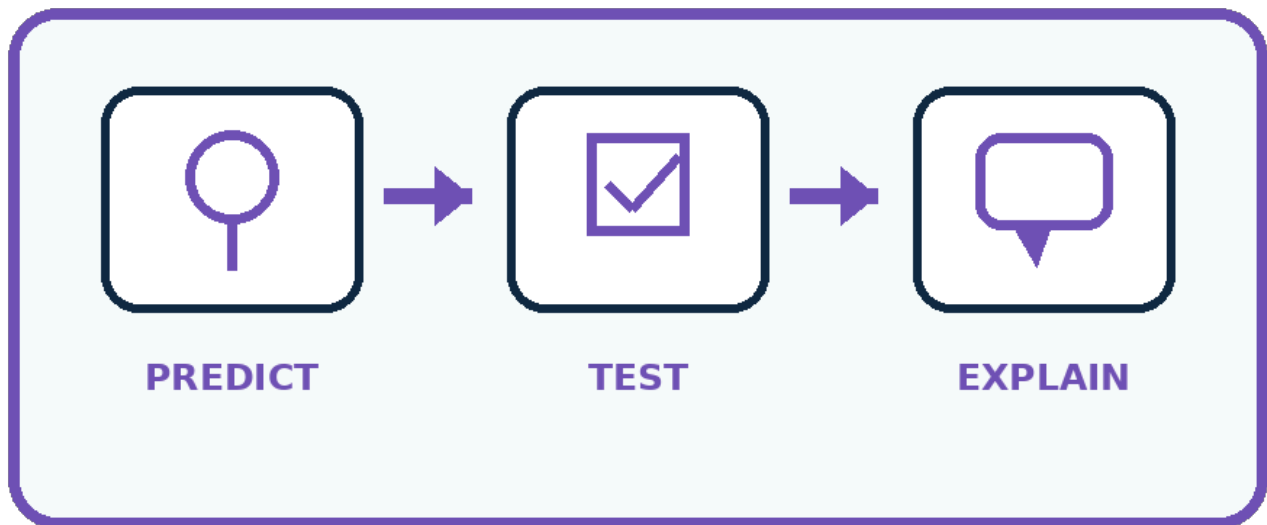
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Need-Finding Interview

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Interview a classmate and write one testable need statement.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

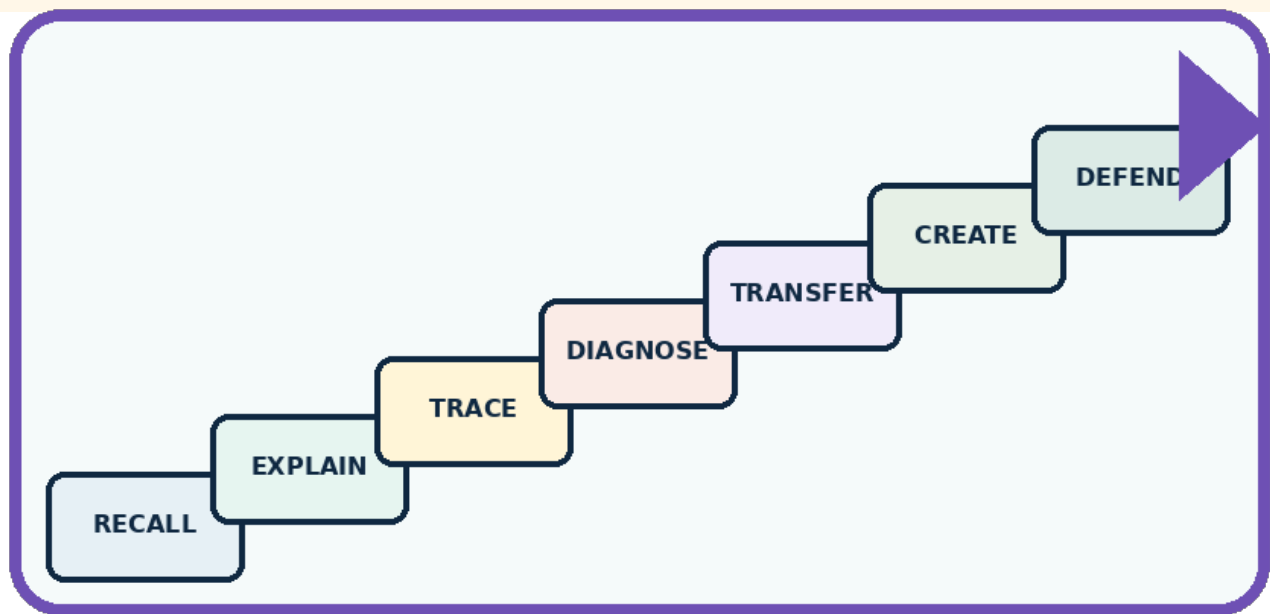
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define user.

EXPLAIN A strong project begins with a specific user and observable need.

TRACE Show the input, process, and output.

DIAGNOSE Repair: A popular technology is proof of a real problem.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

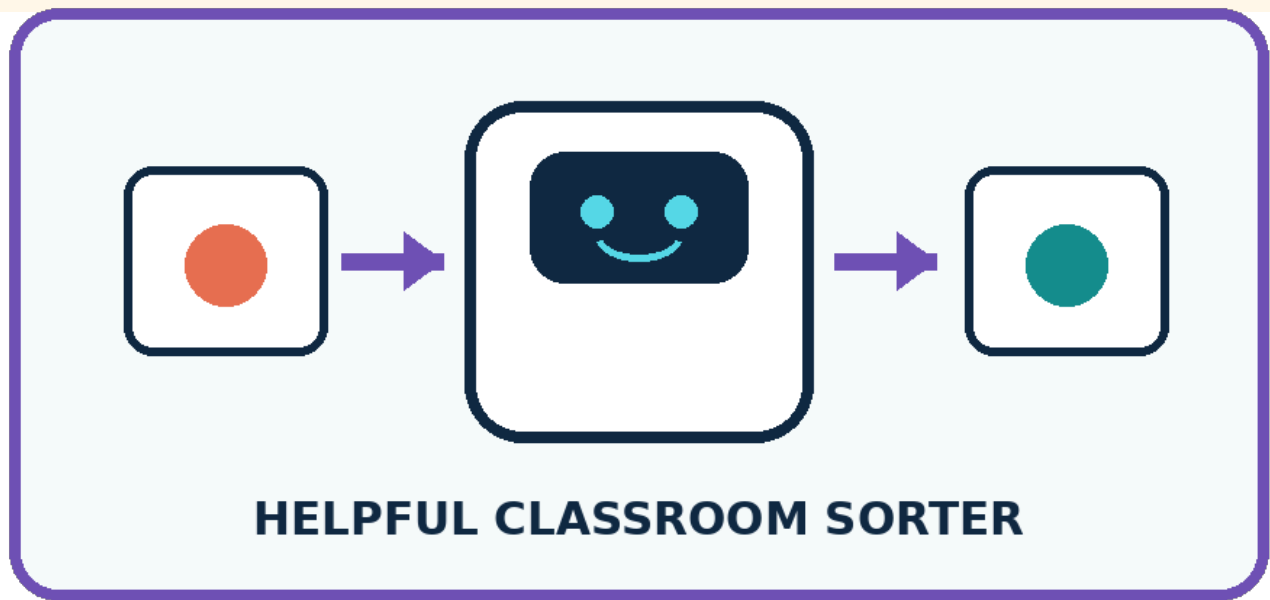
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Choose the Classroom Sorter's real user and measurable need; keep an AI-free solution option open.

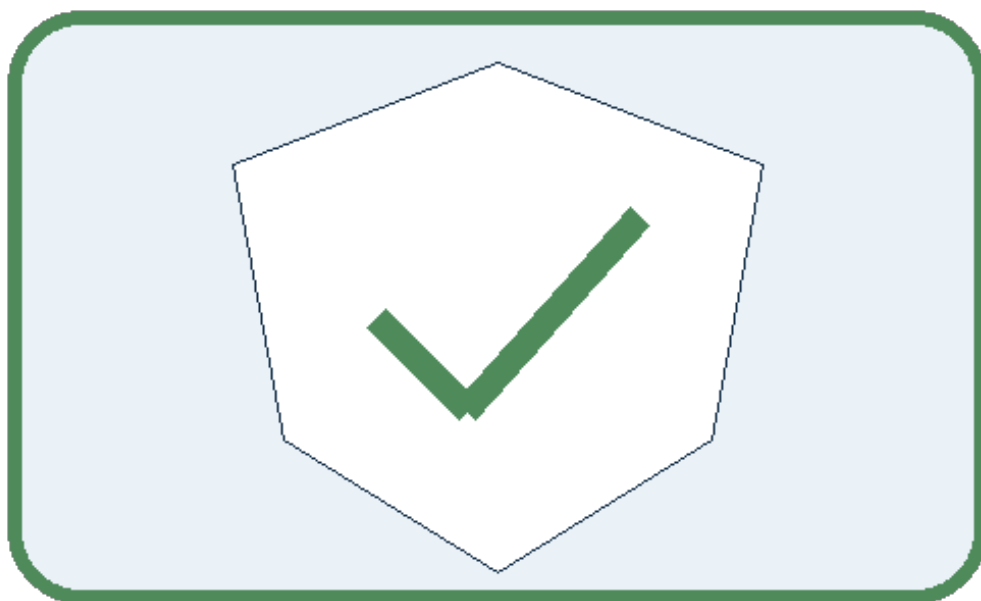
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN Do not collect personal stories unrelated to the design question.

My project decision:

Evidence for your project

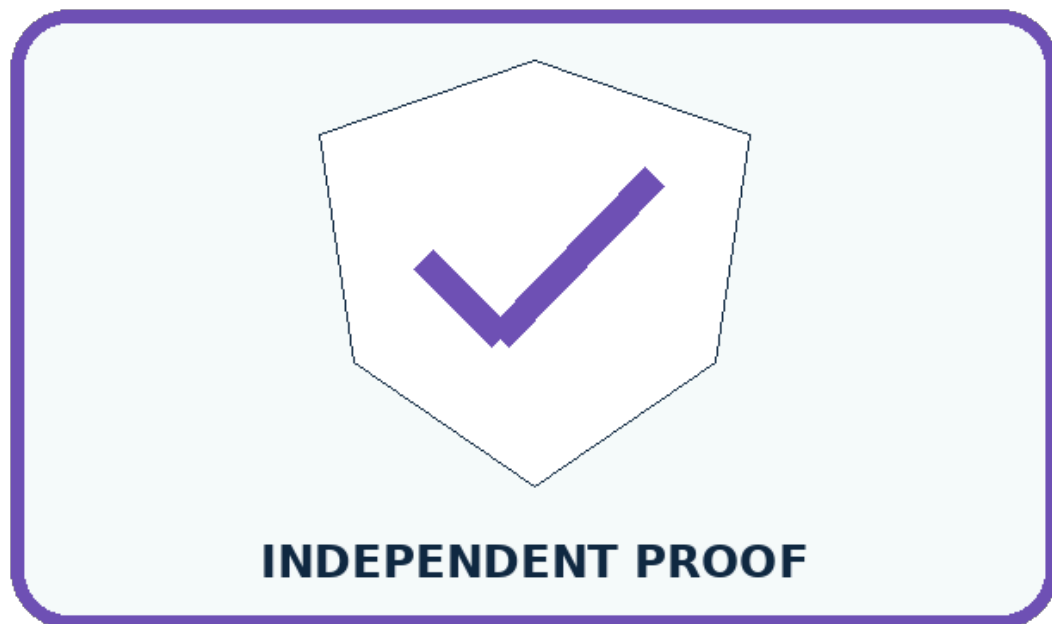
For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



REQUIRED PROOF Interview a classmate and write one testable user-need statement supported by an observation or quote.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK A strong statement names the user, context, observable difficulty, and desired improvement without requiring AI. Protect privacy, avoid leading questions, and distinguish what the user said from your inference.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: Choose Inputs and Outputs.

13. Study a complete case

Week 29 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Shared markers are often lost because tray labels are unclear.

The question

State the need before selecting AI.

Worked explanation

Students need to return markers correctly and quickly. Test clearer labels before proposing a model.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Solve the need before selecting technology

Evidence or stage · Value or result

Evidence or stage: Problem

Value or result: Markers returned to wrong trays

Evidence or stage: Possible cause

Value or result: Unclear tray labels

Evidence or stage: Simple trial

Value or result: Clearer labels

Evidence or stage: Evidence

Value or result: Measure correct and quick returns

Shared markers are often lost because tray labels are unclear.

State the need before selecting AI.

Students need to return markers correctly and quickly. Test clearer labels before proposing a model.

14. Try a new case independently

Independent case: Frame a different problem independently

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do. The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the user difficulty, evidence, desired outcome, necessary information, an excluded action and a simple alternative. Identify one cause or benefit that these counts do not establish.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 29 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Choose the Classroom Sorter's real user and measurable need; keep an AI-free solution option open.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional class loses time finding the correct craft tray.

By the end, I can · How I will show it

By the end, I can: Describe a specific user difficulty using supplied evidence without inventing its cause

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Define a useful outcome and a way to observe it before choosing a solution

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Choose a small project with clear boundaries, permitted information and a simple alternative

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Describe a specific user difficulty using supplied evidence without inventing its cause

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment

for a separate first attempt before teacher feedback.

A fictional class loses time finding the correct craft tray.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use fictional tool requests; do not collect identity or purchase equipment.

1. Bound the project
2. Observe problem
3. Set success test
4. Define need

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Bound the project Use fictional tool requests; do not collect identity or purchase equipment. The mismatch record describes a concrete need. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it. Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

Practice 2: explain the deciding evidence

Changed-case practice: The approved list contains no tray with every requested tool. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. The finder should report no suitable tray rather than guess.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The finder should report no suitable tray rather than guess. Success includes honestly identifying unsatisfied requirements.

Repair a missing building block

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system’s capability, the correctness of its output and permission to take an action. Short practice: Which original proposal is authorised? Why do the other two need separate approval?. Return to this week and explain what you changed.

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied evidence and checkable output requirements. Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Fair for Whom? (ai-1/23). Goal: Compute a correct-out-of-assessed fraction and explain its denominator. Short practice: Which group loses performance under N, and can total improvement erase that fact?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.